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Executive Summary of the Thesis

CCW-SSIM: a Novel Color-Enhanced Structural Similarity Metric applied to Autoencoders for Anomaly Detection

Laurea Magistrale in Computer Science Engineering - Ingegneria Informatica

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1. Introduction

The objective of this work is to integrate structural similarity metrics with deep learning methods for anomaly detection and segmentation. Although the literature has extensively explored this area, existing approaches focus only on **grayscale images**. This is primarily due to the difficulty of crafting a **differential function** as well as the increased dimensionality of colored images, which can be an obstacle in deep learning approaches. Therefore, we introduce a **novel structural similarity metric** that effectively incorporates **color information**.

Autoencoders have emerged as an effective approach in this field. Previous studies, such as those mentioned in [1] and [2], have demonstrated the successful use of *Structure Similarity metrics* (such as SSIM) as loss functions, yielding promising results on grayscale images. To extend those results to color images, we propose a novel Structure Similarity metric called **Color Complex Wavelet Structure Similarity** (**CCW-SSIM**), aimed at extending the high-performance capabilities of the *Complex Wavelet Structure Similarity* to **color images**. Through our experiments, we establish that the new metric surpasses all other loss functions in terms of **AUC-ROC** and **AUC-PR**, as detailed in **Section 5**. In this work, industrial textured datasets are used, but techniques and algorithms presented in this work can be applied to other fields.

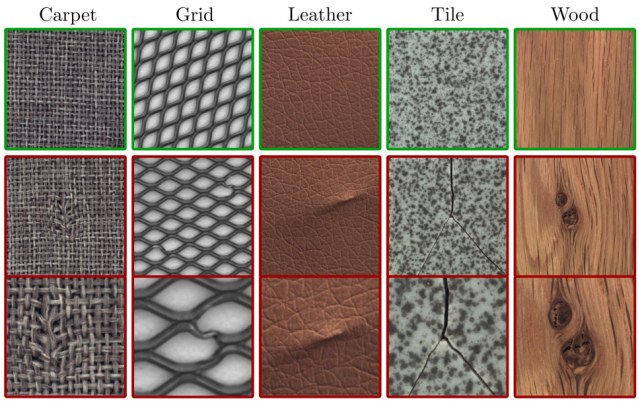


Figure 1: *MVTec Texture dataset [5].*

2. Problem Formulation

Given a set of input images $I = I_1, I_2, ..., I_n$, the problem of anomaly detection and segmentation is to identify and localize anomalous regions within the images.
This can be formalized as a binary segmentation

problem, where the goal is to segment each image I_i into two regions: the anomalous region A_i and the normal region N_i.
Mathematically, this can be expressed as:

$$A_i = \{x \in I_i : f(x) = 1\}$$

$$N_i = \{x \in I_i : f(x) = 0\}$$

where $f : I_i \rightarrow 0, 1$ is a binary function that assigns a label of 1 to anomalous pixels and 0 to normal pixels.
The goal is to optimize the function f, such that it identifies anomalous region in unseen images.

3. Background

3.1. Autoencoder

Autoencoders are artificial neural networks used for unsupervised learning tasks. They consists of an **encoder** network that compresses the input data into a lower-dimensional representation, often referred to as a **latent space**. This compressed representation is then passed through a **decoder** network, which aims to reconstruct the original input data. By comparing the reconstructed output with the input, successfully trained autoencoders can effectively capture and represent the underlying patterns and features of the data.

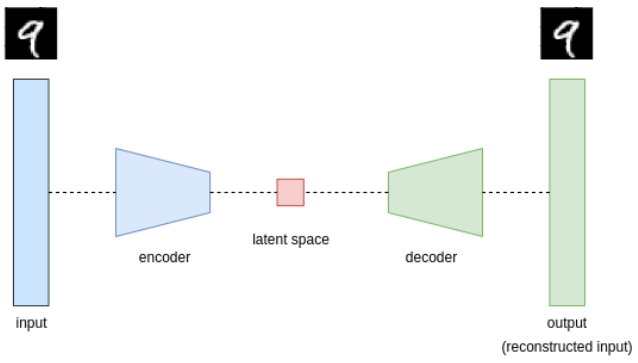


Figure 2: *Autoencoder architecture.*

3.2. Structure Similarity Index Measure

The *Structure Similarity Index Measure* (SSIM)[7] is used for measuring the similarity between two grayscale images. As [1] demonstrated, SSIM can be used as **loss function** for autoencoders, yielding good results. It is defined as:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{1}$$

Where x and y are the images to be compared, μ_x and μ_y are their means, σ_x and σ_y are their standard deviations, σ_{xy} is their covariance, and C_1 and C_2 are constants to stabilize the division. **Equation 1** can be rewritten, underlying its 3 components:

$$SSIM(x, y) = l(x, y) \cdot c(x, y) \cdot s(x, y) \tag{2}$$

Where $l(x, y)$ is the luminance factor, $c(x, y)$ the constrast factor and $s(x, y)$ the structure factor.

Starting from (**2**), a new version of SSIM has been proposed in [4], namely CSSIM [4], adding a color component:

$$CSSIM(x, y) = l(x, y) \cdot c(x, y) \cdot s(x, y) \cdot C_r(x, y) \tag{3}$$

3.3. Color-SSIM (CSSIM)

To compute the color component, $C_r(x, y)$ in **Equation 3**, some processing steps are required. Images are first converted to an opponent color space, then three different spatial filters are convolved to the three different images channels. At the end, images are converted into the **S-CIELAB** color space.

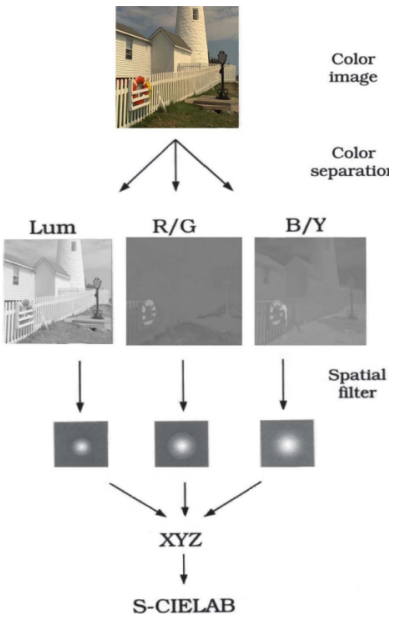


Figure 3: *S-cielab image conversion [4].*

The two-dimensional spatial kernels used are determined by the spatial response of human visual system, to emulate human vision. Parameters are evaluated by physiological and psychological experiments [4]. In S-CIELAB, images are represented with 3 channels:

- L^*: perceptual lightness;
- a^*: red-green opponent value;
- b^*: yellow-blue opponent value.

The color component, $C_r(x, y)$ in **Equation** 3, is defined as:

$$C_r(x, y) = 1 - \frac{1}{k}\Delta E(x, y) \tag{4}$$

where k is an empirically chosen weighting constant and ΔE is the *perceptual color difference*:

$$\Delta E = \sqrt{(L_2^* - L_1^*)^2 + (a_2^* - a_1^*)^2 + (b_2^* - b_1^*)^2} \tag{5}$$

3.4. Complex-Wavelet SSIM

Another limitation of SSIM is its high sensitivity to small rotations and traslations. To address such limitations, an extension of SSIM to the complex wavelet domain has been proposed in [6]. This variant is insensitive to small differences, that are usually caused by nuisance factors. We define the *Complex Steerable Pyramid Decomposition* as a multi-scale and multi-orientation image representation that decomposes an image into a set of band-pass subbands with different scales and orientations.

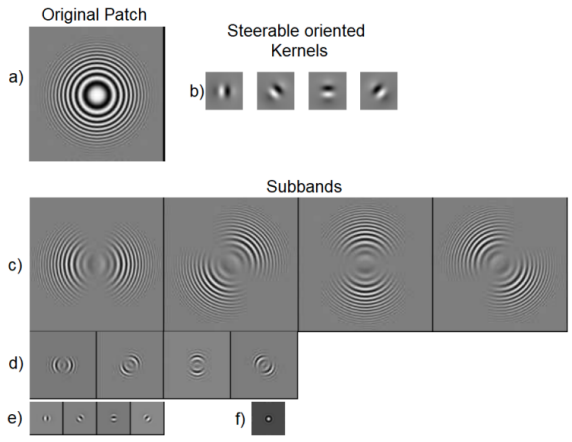


Figure 4: *5 scales 4 orientations steerable pyramid. Image from [3].*

In order to compare images using the CW-SSIM index, a **complex 6-scale, 6-orientation steerable pyramid decomposition** is used to decompose the images. The decomposition process involves filtering the image with a set of oriented filters at multiple scales. The outputs of these filters are then combined to create a set of subbands at each scale. The resulting subbands are further filtered and downsampled to create the next level of the pyramid. This process is repeated until the desired number of levels is reached. In the complex wavelet transform domain, we define two sets of coefficients $\mathbf{c}_x = \{c_{x,i} \mid i = 1, \ldots, N\}$ and $\mathbf{c}_y = \{c_{y,i} \mid i = 1, \ldots, N\}$ at the same spatial location in the same wavelet subbands of the two images being compared, respectively. Using these sets of coefficients, we can calculate the CW-SSIM index, which provides a measure of the similarity between the two images.

$$\mathrm{CW}\left(\mathbf{c}_x, \mathbf{c}_y\right) = \frac{2\left|\sum_{i=1}^{N} c_{x,i} c_{y,i}^*\right| + K}{\sum_{i=1}^{N} |c_{x,i}|^2 + \sum_{i=1}^{N} |c_{y,i}|^2 + K} \tag{6}$$

Where c^* denotes the complex conjugate of c and K is a small positive constant, added to avoid weird behaviours when denominator approaches zero. The CW-SSIM indices are then calculated locally by sliding a **7x7** window across each wavelet subband. To obtain a single value similarity index, the CW-SSIM is averaged over all the windows:

$$\mathcal{L}_{cw}(x, y) = 1 - \frac{1}{N}\sum_{i=1}^{N} \mathrm{CW}\left(c_{x,i}, c_{y,i}\right) \tag{7}$$

3.5. Deep Autoencoders for Anomaly Detection in Textured Images using CW-SSIM

This work is an extension of [2], where the CW-SSIM was proposed as loss function to tackle the problem of anomaly detection with autoencoders, with good results on grayscale images. We propose a possible extension of [2] to **colored images**. As stated in [2], the color components of images are often very important in the context of anomaly detection.

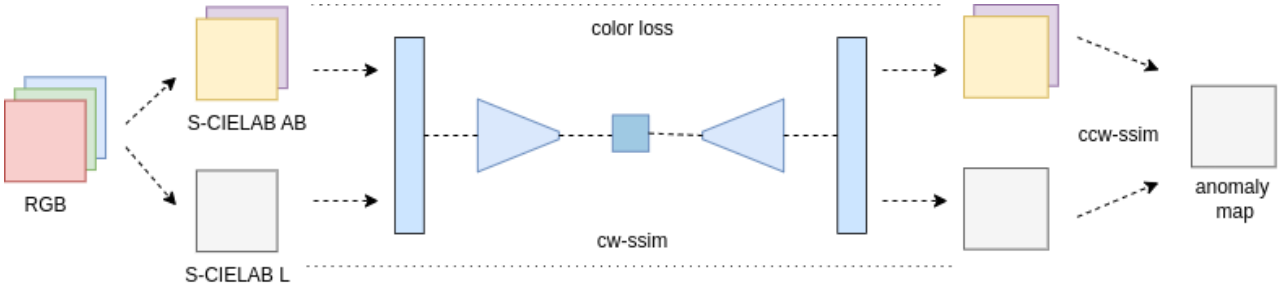


Figure 5: *CCW-SSIM as loss function for autoencoders.*

4. Proposed Solution

As shown in [2], the CW-SSIM loss function is highly effective in autoencoder-based anomaly detection applications, specifically in the domain of grayscale images. However, this function does not readily extend to the detection of color anomalies. To address this limitation, a **novel image similarity metric** has been proposed in this study, with the objective to retain the great performance of CW-SSIM on grayscale, while enabling the detection of color anomalies.Following the idea of [4], we added the color component $C_r(x, y)$, as shown in **(5)**, to the CW-SSIM, with some small modifications. In detail, the perceptual color difference shown in **(3)**, becomes:

$$\Delta E = \sqrt{(a_2^* - a_1^*)^2 + (b_2^* - b_1^*)^2} \tag{8}$$

In the **LABs** color spaces, contrary to other spaces like RGB, there is a clear distinction between color components and grayscale component. The whole color information contained in the image is stored in the a^* and b^* planes. Hence, we compute the color component of the similarity metrics as shown in **Equation 8**. Now we define the new color component $C_r(x, y)$ as:

$$C_r(x, y) = 1 - \frac{1}{k}\Delta E(x, y) \tag{9}$$

where ΔE is the color fidelity value computed in **Equation 8**, k is a weighting constant to be defined.

4.1. CCW-SSIM

We can now define the new *Color Complex Wavelet Structure Similarity Index*, **CCW-SSIM** as:

$$\text{CCW-SSIM}(x, y) = \text{CW-SSIM}(x, y) \cdot C_r(x, y) \tag{10}$$

where x and y are the the two images (or patches) to be compared. Starting from **Equation 10**, which represents a *similarity*, we can compute the corresponding loss:

$$CCWSSIM_{loss}(x, y) = 1 - \text{CCW-SSIM}(x, y) \tag{11}$$

To identify anomalies within an image, we need to assign a loss value to each pixel of the image. This is achieved by computing **Equation 11** within sliding windows. By doing so, the loss value of the window centered on a given pixel is assigned to that particular pixel. Through tuning process, a window size of **7x7** has been found to provide the most optimal results. The CCW-SSIM can readily be used in the training loop since it is a variant of the CW-SSIM with simple additional operations.

4.2. Rationale

In this study, we chose to employ the S-CIELAB color space for two crucial reasons. Firstly, S-CIELAB is **perceptually uniform** color space, where a specific numerical variation corresponds to a corresponding perceived change in color. This characteristic enables more meaningful computation of color differences. Secondly, as previously described, the LAB color spaces maintain a clear distinction between color and grayscale components. The latter properties of LAB spaces allows us to consider only **color features** when computing the color component in our loss function. By doing so, the L plane is only used to compute the CW-SSIM component of the loss function. This approach enables us to maintain the good performance of CW-SSIM function while extending it to include colors.
Another advantage of this modification can be noticed at training time. Since the two loss

components are computed on different channels, they are not in competition, making the selection of the parameter K in **Equation 9** very easy. In other words, its value needs to be tuned based solely on the learning rate, without regard to the relative weight we assign to the structural and color components. That is because it is not important in the reconstruction phase but only for anomaly map computation.

4.3. Color-Structural Equilibrium

We followed a simple strategy to achieve a balanced weighting between the structural (CW-SSIM) and color components. By computing some structural anomaly maps, we determined the average anomaly score per pixel. Subsequently, we fine-tuned the color parameter (k in **Equation 9**) to match the average anomaly score of the color anomaly map. Experimental results demonstrated that a smaller value led to superior results, confirming the dominant significance of the structural component over the color component for anomaly detection. Hence, we slightly reduced the parameter value.

4.4. Autoencoder Architecture

We implemented the same autoencoder architecture presented in [2], with a slightly bigger latent-space. To obtain the most optimal result, a patchwise prediction is done. Test images are cropped in overlapping patches of size **256x256**, with **stride 16**, before being fed to the autoencoder. Values are then averaged on overlapping pixel. This procedure increases accuracy but is time-expensive, that is why we decided to develope a **fully convolutional** autoencoder. The fully-convolutional architecture allows us to feed the autoencoder with full images at prediction time.

4.5. Experiments

The training phase has been done with the **ADAM** optimizer with **50000** defect-free **128x128** patches. The starting learning rate was set to $1e^{-3}$, halving it every 15 epochs.
The following tables report the results, in terms of **AUC-ROC** and **AUC-PR**, both normalized up to (**FPR=0.3**) of different loss functions used with autoencoders:

Dataset	mse	ssim	cwssim	ccwssim
Leather	0.7161	0.8879	0.9734	**0.982**
Carpet	0.5678	0.8387	0.8629	**0.8946**
Tile	0.5998	0.7619	0.8096	**0.8738**
Wood	0.6875	0.7179	0.7219	**0.8004**

Table 1: *Normalized AUC-ROC on MVTec of various loss functions.*

Dataset	mse	ssim	cwssim	ccwssim
Leather	0.5735	0.7421	0.9563	**0.9736**
Carpet	0.232	0.7393	0.7549	**0.8439**
Tile	0.3917	0.6490	0.7547	**0.7734**
Wood	0.558	0.6008	0.5472	**0.7036**

Table 2: *Normalized AUC-PR on MVTec of various loss functions.*

The **CCW-SSIM** trained autoencoder yields slighlty better but consistent results, compared to other loss functions. This demonstrates the importance of colors in anomaly detection, as predicted in [2]. For the vast majority of samples the grayscale **structural component** is enough to detect anomalies, as shown in **Figure 6**.

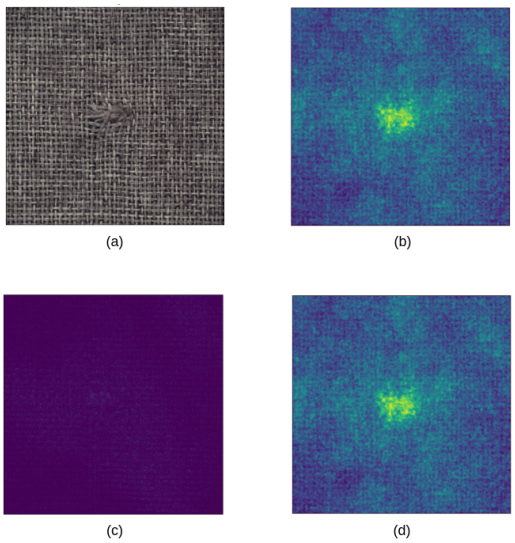


Figure 6: *Example of anomaly map. (a): original image. (b): CW-SSIM anomaly component. (c): color anomaly component. (d): CCW-SSIM anomaly map.*

This is why the performance reported before are

only slightly better than those obtained with the grayscale CW-SSIM. However, some anomalies are not detectable on grayscale. In **Figure 7** we reported an example where the color component is crucial.

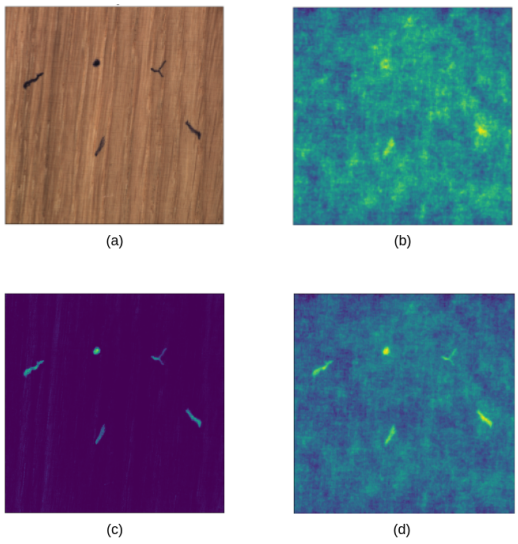


Figure 7: *Example of anomaly map. (a): original image. (b): CW-SSIM anomaly component. (c): color anomaly component. (d): CCW-SSIM anomaly map.*

In **Figure 7**, the limitation of grayscale CW-SSIM is evident. Yet, including the metric of the **color component** enables the detection of anomalous pixels.

As previously explained, the fully convolutional architecture allow us to predict entire images at once. The following table shows the difference in term of prediction time between patchwise and imagewise prediction:

Imagewise vs Patchwise Prediction

	Imagewise	**Patchwise**
Time	23.54s	2749.5s

Table 3: *Autoencoder prediction time on the Carpet MVTec Dataset.*

As indicated in **Table 3**, imagewise prediction leads to a significant acceleration of the model. However, this improvement comes at the cost of approximately a 10% performance degradation.

5. Conclusions

In this work we proposed a new **color-structure similarity metric** and evaluates it as loss function for autoencoders in the context of anomaly segmentation. Our newly developed metric, when used as a loss function, shows promising results. While it may not yet reach the performance of sophisticated state-of-the-art methods, it consistently outperforms all other tested loss functions when employed with the same architecture. This encourages further exploration and refinement of our metric for even better performance in the future.

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